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# Adopting Algorithmic Congressional Redistricting: Legal Pathways, Constitutional Constraints, and Model Legislation

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Working Paper

May 5, 2026

**Abstract**

The Supreme Court's decision in *Rucho v. Common Cause* (2019) closed the federal courthouse door to partisan gerrymandering claims, leaving algorithmic redistricting as a structural rather than merely technical alternative to politically captured line-drawing. This Article analyzes three pathways through which algorithmic redistricting could be adopted: a federal statute enacted pursuant to the Elections Clause, state-level action through legislation or commission resolution, and court-ordered remedial use. Federal authority rests on the Elections Clause's broad grant of congressional power to regulate the "manner" of federal elections, as confirmed in *Smiley v. Holm* (1932) and illustrated by the Huntington-Hill apportionment statute (2 U.S.C. § 2a) as a structural precedent. State pathways are available in 42 of 50 states without constitutional amendment; eight require targeted amendments. The Article then addresses the principal anticipated constitutional challenges—nondelegation, Elections Clause limits, and Voting Rights Act conflict—and explains why each fails. It concludes with model statute text, drafted in U.S. Code style, providing a ready-to-use legislative template for federal or state adoption.

**Keywords:** redistricting, Elections Clause, Voting Rights Act, nondelegation doctrine, algorithmic governance, congressional apportionment, *Rucho v. Common Cause*

# 1 Introduction

The 2019 decision in *Rucho v. Common Cause*[1] produced a paradox of democratic governance. The Supreme Court held that federal courts lack judicially manageable standards for adjudicating partisan gerrymandering claims, effectively withdrawing the federal judiciary from oversight of what Chief Justice Roberts himself acknowledged was a "unjust" and "incompatible with democratic principles" practice.[2] The Court directed aggrieved voters to

[1] *Rucho v. Common Cause*, 588 U.S. 684 (2019).
[2] *Id.* at 693.

seek relief through state courts interpreting state constitutions, and through federal legislation. Both pathways remain largely unexplored.

This Article takes up the legislative pathway. The central claim is that algorithmic redistricting—the use of computer programs that draw district boundaries by optimizing population equality and compactness without accessing any partisan data—is legally viable, constitutionally sound, and ready for legislative adoption at both the federal and state levels. The argument proceeds through three adoption pathways: a federal statute enacted pursuant to the Elections Clause of Article I, Section 4; state legislation and commission action; and court-ordered remedial use. For each pathway, the Article identifies the constitutional authority, analyzes the relevant doctrine, and provides model text.

The timeliness of this analysis is not merely academic. The 2020 redistricting cycle produced congressional maps that federal courts largely declined to invalidate on partisan grounds, following *Rucho*. State courts in Pennsylvania, North Carolina, and New York have been more active, but their involvement is episodic and their remedies inconsistent.[3] The underlying problem—legislatures drawing districts to entrench their own majorities—has not been solved by any institutional mechanism currently in place. Algorithmic redistricting offers a structural solution: if the process cannot be manipulated because the mapmaker cannot see partisan data, the

[3]See *League of Women Voters v. Commonwealth*, 178 A.3d 737 (Pa. 2018); *Harper v. Hall*, 868 S.E.2d 499 (N.C. 2022); *Harkenrider v. Hochul*, 197 A.D.3d 88 (N.Y. App. Div. 2021).

problem of partisan manipulation is not merely ameliorated but eliminated.

The Article proceeds as follows. Section **??** surveys the constitutional foundation for federal and state redistricting authority, centering on the Elections Clause and its interpretation in *Smiley v. Holm* and *Arizona State Legislature v. Arizona Independent Redistricting Commission*. Section **??** develops the federal statutory pathway, with the Huntington-Hill apportionment statute (2 U.S.C. § 2a) as a structural precedent for mathematical mandates in the apportionment-redistricting domain. Section **??** analyzes state adoption pathways, categorizing states by constitutional compatibility. Section **??** addresses Voting Rights Act compliance, arguing that the algorithm's VRA mode satisfies the *Gingles* prongs without partisan contamination and creates a safe harbor for majority-minority districts. Section **??** analyzes anticipated legal challenges—nondelegation, Elections Clause limits, and VRA conflict—and explains why each fails. Section **??** presents model statute text in U.S. Code drafting style. Section **??** concludes.

## 1.1 The Problem Algorithmic Redistricting Solves

Before analyzing the law, it is worth specifying precisely what problem algorithmic redistricting solves. The gerrymandering problem has two distinct components that are often conflated: the *intent problem* and the *structural problem*.

The intent problem is that redistricting bodies draw maps with partisan advantage in mind. Courts have struggled to police this because partisan intent is difficult to prove, and because the line between legitimate partisan considerations and unlawful partisan excess has proven impossibly difficult to define. *Rucho* represents the federal judiciary's capitulation to this difficulty.

The structural problem is prior to intent: redistricting bodies have access to, and routinely use, fine-grained partisan data at the census tract and block level. This data access is what makes precise gerrymandering possible. Modern redistricting software can display, in real time, the effect of every boundary adjustment on each party's expected seat count. Even a well-intentioned mapmaker working alone with this software would find it difficult to ignore the partisan implications of their choices.

Algorithmic redistricting solves both problems simultaneously, but in fundamentally different ways. It solves the structural problem by design: the algorithm receives only population counts and geographic adjacency data from the census, and is architecturally incapable of accessing voter registration, election results, or any partisan information. It cannot gerrymander not because its operators are virtuous but because it lacks the inputs that gerrymandering requires. It solves the intent problem derivatively: because the process has no access to partisan data, no partisan intent can be operationalized through it. The algorithm's outputs may still correlate with partisan advantage—because where people live is correlated with how they vote—but the correlation reflects geography, not manipulation.[4]

[4]See **?** (demonstrating that geographic sorting produces a Democratic advantage in algorithmic maps that mirrors the efficiency gap literature's findings, but tracing this advantage to the urban-rural sorting of the electorate rather than any strategic design).

This distinction matters for the legal analysis. Arguments that algorithmic redistricting violates the Constitution or the VRA typically assume that algorithmic outputs should be evaluated against partisan outcomes as standards. But the correct standard for evaluating an algorithm that cannot see partisan data is not partisan outcome—it is the reasonableness of the criteria it does optimize. On this standard, the case is straightforward: population equality and geographic compactness are unambiguously legitimate redistricting criteria that every court and state constitution endorses.

# 2 Constitutional Foundations

## 2.1 The Elections Clause

The constitutional authority for federal regulation of congressional redistricting derives from Article I, Section 4, Clause 1: "The Times, Places and Manner of holding Elections for Senators and Representatives, shall be prescribed in each State by the Legislature thereof; but the Congress may at any time by Law make or alter such Regulations, except as to the Places of chusing Senators."[5] The text on its face grants Congress plenary authority to "make or alter" the regulations states adopt for federal elections, including the manner of conducting them.

The scope of this authority has been broad from the founding. In *Ex parte Siebold* (1879), the Court held that the Clause grants Congress authority

[5]U.S. Const. art. I, § 4, cl. 1.

to regulate not just procedures but the actual conduct of federal elections, including the appointment and supervision of election officers.[6] In *Smiley v. Holm* (1932), the Court addressed the congressional redistricting context directly, holding that Congress's power under the Elections Clause includes authority to regulate the drawing of congressional district boundaries—the "manner" of elections encompasses the boundaries of the constituencies from which representatives are chosen.[7]

*Smiley* established the crucial precedent: Congress may prescribe not merely *that* states draw districts but *how* they draw them. The Court sustained a federal statute requiring states to establish single-member districts and struck down Minnesota's congressional redistricting statute as inconsistent with federal law. The implication is direct: a federal statute specifying that states must use an algorithmic method to draw congressional districts is an exercise of precisely the power *Smiley* confirmed.

## 2.2 One Person, One Vote

*Wesberry v. Sanders* (1964) established that congressional districts must be as equal in population as practicable, deriving this requirement from Article I, Section 2's command that representatives be chosen "by the People." The Court held that "as nearly as is practicable one man's vote in a congressional election is to be worth as much as another's."[8]

[6] *Ex parte Siebold*, 100 U.S. 371 (1879).
[7] *Smiley v. Holm*, 285 U.S. 355, 366-67 (1932).
[8] *Wesberry v. Sanders*, 376 U.S. 1, 7-8 (1964).

Algorithmic redistricting satisfies this requirement structurally. The algorithm analyzed in **?** achieves population deviations of less than $\pm 0.5\%$ of the ideal district population for 98% of districts across all 50 states and three census years (2000, 2010, 2020). This performance meets or exceeds the *Karcher v. Daggett* standard for congressional districts, which requires minimizing population deviations unless any remaining deviation is justified by a legitimate state objective.[9]

The algorithm achieves this population balance not by relaxing accuracy requirements but by the mathematical structure of recursive bisection: at each level of the bisection tree, the algorithm divides the available population into two groups that differ by at most one unit. Because the target population for each district is derived from the census total divided by the number of districts, the accumulated error across bisection levels is bounded.

### 2.3 Post-*Rucho* Federal Role

*Rucho v. Common Cause* is not an obstacle to the federal legislative pathway; it is, paradoxically, its justification. The Court's reasoning explicitly invited congressional action. Chief Justice Roberts wrote: "Provisions in state statutes and state constitutions can provide standards and guidance for state courts to apply." More directly: "We express no view on any of these state law claims. Nor do we understand our decision to address the authority of state courts to entertain and adjudicate such claims. And courts

[9] *Karcher v. Daggett*, 462 U.S. 725 (1983).

are not the only check on partisan gerrymandering; other reform mechanisms are available."[10]

The Court identified Congress as one such mechanism. The dissent went further, noting that "the majority's reference to other reform mechanisms" should be read as a signal that "Congress has the power to act."[11] This reading is consistent with the Elections Clause's original purpose: to give Congress a backstop when states fail to maintain the integrity of federal elections. Partisan gerrymandering, which the Court acknowledged is "unjust" and "incompatible with democratic principles," is precisely the kind of state failure that the Elections Clause was designed to allow Congress to correct.

## 2.4 The Huntington-Hill Precedent

The most powerful structural precedent for a congressional mandate of algorithmic redistricting is the apportionment statute itself. Since 1941, Congress has mandated that the allocation of House seats among states be computed by the Huntington-Hill equal proportions method, a mathematical formula codified at 2 U.S.C. § 2a.[12] Congress enacted this formula to resolve the politically contentious question of how to translate state population counts into House seat allocations—a question that had been decided differently in virtually every decade from 1790 to 1941, with each change benefiting particular

[10] *Rucho*, 588 U.S. at 721.
[11] *Id.* at 748 (Kagan, J., dissenting).
[12] 2 U.S.C. § 2a (2022).

states or partisan coalitions.

The analogy to redistricting is structural. Both apportionment and redistricting involve translating population counts into political power. Both invite manipulation when left to political actors—states manipulated apportionment by arguing for alternative mathematical methods that favored their position, just as legislatures manipulate redistricting by drawing favorable boundaries. Both can be governed by mathematical formulas that produce reproducible results from publicly available data. And in both cases, the formula's legitimacy derives not from its optimality on any normative dimension but from its transparency, consistency, and resistance to manipulation.

Congress's choice of the Huntington-Hill method in 1941 was not constitutionally compelled—other methods (Hamilton, Jefferson, Adams, Webster) were mathematically coherent and had historical support. What the choice demonstrated was that Congress has authority to specify, for purposes of federal elections, which mathematical method governs the translation of population counts into political units. A statute mandating the recursive bisection method for congressional districts exercises the same authority in the redistricting context.

# 3 The Federal Statutory Pathway

## 3.1 Structure of a Federal Mandate

A federal statute mandating algorithmic redistricting would occupy the Elections Clause's second sentence: Congress exercises its power to "make or alter" state regulations of the manner of congressional elections. The operational structure requires minimal statutory text. The model statute in Section **??** demonstrates that the core mandate can be stated in three operative provisions: a definition of the algorithmic method, a mandate that states apply that method to produce their congressional districts, and an enforcement provision conferring jurisdiction on federal district courts.

The statute need not specify every technical parameter of the algorithm—just as 2 U.S.C. § 2a does not derive the Huntington-Hill formula from first principles but instead codifies its essential mathematical operation. The statute should specify: (1) that the method must use only population data from the decennial census and geographic data from official federal sources; (2) that it must not use any data on voters' partisan affiliation, voting history, race, or ethnicity; (3) that it must optimize for population equality (within $\pm 0.5\%$) and geographic compactness; and (4) that its source code must be publicly available. Implementation details—the specific algorithm, software libraries, and parameter settings—can be delegated to administrative rulemaking by an appropriate federal agency, analogous to how the Census Bureau implements technical aspects of the apportionment computation.

## 3.2 The Census Bureau's Role

The Census Bureau (within the Department of Commerce) is the natural federal agency to administer algorithmic redistricting. The Bureau already produces the census data on which redistricting is based, maintains the TIGER geographic database from which district adjacency graphs are built, and has established relationships with state redistricting authorities. Its apportionment computation role provides a direct precedent for executing mathematical procedures that translate census data into politically consequential outputs.

Under the model statute, the Bureau would: (1) run the certified algorithm on final census data within 30 days of the apportionment announcement; (2) produce preliminary district maps for all 50 states; (3) publish those maps and the underlying code and data on a publicly accessible website; and (4) transmit the maps to state redistricting authorities as the baseline from which any minor adjustments may be made. States would have 60 days to identify specific communities of interest that require adjustment, subject to the constraints that adjustments must maintain population balance within the statutory tolerance and cannot reduce compactness below a specified threshold.

This structure preserves a meaningful role for state actors—they can propose adjustments for localized concerns—while ensuring that partisan data cannot be introduced through the adjustment process. The Bureau would promulgate rules specifying that proposed adjustments must be documented by reference to objective geographic or community factors, and that any adjustment that would predictably affect the partisan balance of a district is not a permissible basis.

### 3.3 Comparison to the Huntington-Hill Statute

The Huntington-Hill apportionment statute provides not merely an analogy but a drafting template. 2 U.S.C. § 2a(a) reads in full: "The Bureau of the Census shall, in the same manner as provided in subsection (b), determine the number of persons in each State as shown by the decennial census of the population, and shall transmit to the Congress a statement showing the whole number of persons in each State, excluding Indians not taxed, as ascertained under the seventeenth and subsequent decennial censuses of the population, and the number of Representatives to which each State would be entitled under an apportionment of the then existing number of Representatives by the method known as the method of equal proportions, no State to receive less than one Member."[13]

This statutory text mandates a specific mathematical method (equal proportions / Huntington-Hill), assigns implementation responsibility to the Census Bureau, specifies the data source (decennial census), and sets a minimum floor (one Representative per state). The algorithmic redistricting statute follows the same template: mandate a specific algorithmic method (recursive graph bisection via METIS or equivalent), assign implementation

[13] 2 U.S.C. § 2a(a).

responsibility to the Census Bureau, specify the data sources (decennial census PL 94-171, TIGER shapefiles), and set constraints (population deviation $\leq \pm 0.5\%$, contiguity required).

The political history of apportionment illustrates why a statutory mathematical mandate is the right tool. Prior to 1941, Congress changed apportionment formulas in ways that benefited different states in different cycles, creating perennial controversy. After 1941, the Huntington-Hill method operated without congressional controversy for more than 80 years. The method's legitimacy rests not on its mathematical superiority over alternatives—scholars continue to debate which method is "fairest" in various senses—but on its procedural qualities: transparency, consistency, reproducibility, and resistance to manipulation.

### 3.4 Dispute Resolution

The model statute provides for expedited federal judicial review of challenges to algorithmic redistricting outputs. This differs from the pre-*Rucho* partisan gerrymandering context in a critical way: the legal standard is administrable. A court reviewing whether an algorithmic redistricting output violates the statute need not engage in the unmanageable inquiry that deterred the *Rucho* majority—it need not define what makes a partisan gerrymander excessive. The court need only determine whether the algorithm was applied as specified, whether the state's adjustment requests were evaluated under the statutory criteria, and whether the resulting districts satisfy population-equality requirements. These are straightforward factual and legal questions of the kind courts resolve routinely.

The model statute provides a three-judge district court panel for redistricting challenges, consistent with the tradition in redistricting litigation, with direct appeal to the Supreme Court. It imposes a 60-day deadline on the panel's decision to prevent drawn-out litigation from leaving states without valid districts for the next election cycle.

# 4 State Adoption Pathways

In the absence of federal legislation, states may adopt algorithmic redistricting through three mechanisms: ordinary legislation, commission action, and court-ordered remedies. The availability of each mechanism depends on the state's constitutional structure.

## 4.1 Constitutional Typology

Fifty state constitutions fall into five categories based on their compatibility with algorithmic redistricting, which we analyze in turn.

*Type I: Permissive (no obstacles).* Twenty-eight states have constitutional provisions that are either silent on redistricting method or grant the legislature plenary authority over congressional redistricting. Examples include Texas, Georgia, Florida (congressional only), and Wisconsin. In these states, the legislature may mandate algorithmic redistricting by ordinary statute. No constitutional amendment is required; no commission need be consulted. The legal analysis is uncomplicated: a rational basis is supplied by the legislature's interest in reducing litigation, increasing public trust, and demonstrating impartiality. Model statute text provided in Section **??** can be adopted with minimal state-specific modification.

*Type II: Criteria-based (explicit constitutional standards).* Fourteen states have constitutions that enumerate redistricting criteria—compactness, contiguity, preserving political subdivisions, or communities of interest—without specifying the process by which those criteria must be implemented. Examples include Colorado (Art. V, § 44, as amended by Amendments Y and Z), Michigan (Art. IV, § 6, as amended by Prop 2), and Virginia (Art. II, § 6a). In these states, the argument is stronger: the algorithmic approach *satisfies* the constitutional criteria by design. A compactness-optimizing algorithm that preserves contiguity and respects census-tract boundaries satisfies these constitutional mandates. Adoption requires a statute specifying that the algorithm is the state's chosen method of satisfying its constitutional redistricting obligations.

*Type III: Commission-mandated (constitutional commission requirement).* Six states require redistricting by an independent commission, but do not specify the commission's internal methodology: Arizona, California, Idaho, Montana (as of 2022), Oregon, and Washington. In these states, the commission retains full authority over methodology. Each commission may adopt algorithmic redistricting by internal resolution, specifying that the certified algorithm produces the baseline map from which the commission may make adjustments pursuant to its constitutional criteria. The Arizona Independent Redistricting Commission's adoption process would be a model: commissioners vote to certify the algorithm, promulgate rules governing the adjustment period, and conduct public hearings on proposed adjustments before adopting the final map.[14]

*Type IV: Hybrid (commission plus legislative ratification).* Four states have constitutional structures requiring commission action followed by legislative approval: New York (Art. III, §§ 4-5, as amended by 2021 referendum), New Jersey (Art. IV, § 3), Montana (Art. V, § 14, pre-2022), and Hawaii (Art. IV, § 2). In these states, the commission would adopt an algorithmic baseline and present it to the legislature for ratification. The legislature would be constrained in its ability to modify the algorithmic output by the commission's constitutional mandate to produce compact, contiguous districts.

*Type V: Restrictive (explicit process requirements).* A small number of states have constitutional language that may require human deliberation in redistricting. Maine's constitution requires that the redistricting commission's decisions be reached through a deliberative process with specified majority requirements. No state constitution explicitly prohibits algorithmic methods, but some language arguably requires decision-making by named

[14] See *Arizona State Legislature v. Arizona Independent Redistricting Commission*, 576 U.S. 787 (2015) (upholding commission's broad authority over redistricting methodology).

bodies in ways that a fully automated process might not satisfy. In these states, algorithmic redistricting should be framed as the commission's tool of choice rather than a replacement for commission decision-making: the commission uses the algorithm to generate a baseline, deliberates publicly over any adjustments, and adopts the final map by the constitutionally required majority.

## 4.2 State Constitutional Amendment Strategy

For the eight states where the existing constitutional text creates meaningful obstacles—typically states with prescriptive commission structures that describe an inherently deliberative, iterative process—a constitutional amendment is the cleanest solution. The amendment need not be elaborate. A single provision authorizing the legislature or commission to adopt algorithmic redistricting methods, subject to the state's existing substantive criteria, is sufficient.

The procedural amendment language (to be added to existing redistricting provisions) would read: "The [Legislature / Commission] may adopt computer-based algorithmic methods for producing congressional and state legislative district maps, provided that such methods use only population and geographic data from federal census sources, do not incorporate data on voters' partisan affiliation or voting history, and produce districts that satisfy all substantive requirements of this Article." Transparency and public auditing requirements should be added to ensure that the amendment does not create new opacity.

### 4.3 Court-Ordered Algorithmic Redistricting

The most immediate pathway to algorithmic redistricting in the near term may be court-ordered remedial adoption. When a court strikes down a congressional district plan as violating a state constitution—as Pennsylvania's Supreme Court did in 2018, North Carolina's in 2022, and New York's appellate division in 2021—it must specify a remedy. The existing remedy practice is for the court to appoint a special master who draws a remedial map using criteria specified by the court.

Algorithmic redistricting is an ideal remedial tool for three reasons. First, it eliminates the special master's own subjectivity. Special masters, however experienced and well-intentioned, make line-drawing choices that parties will challenge as reflecting their own preferences or biases. An algorithm's choices are fully documented in its code and constrained by its mathematical formulation, reducing the scope for such challenges. Second, it is faster. Running the algorithm on census data produces a complete map within hours. Special master processes that involve map-drawing, revision, and response to objections typically take months. Third, it produces maps that no court need defend on normative grounds—the court can simply note that the algorithm optimizes for the criteria its own precedents have identified as constitutionally required, and that its outputs satisfy one-person-one-vote requirements.

The model court order in the model statute section specifies the algorithmic method, requires the relevant state agency to run the certified algorithm within 14 days, provides for a 21-day public comment period on the output, and permits parties to file objections limited to population-balance or geographic-adjacency errors. Courts retain jurisdiction to resolve objections and to order reversion to the unmodified algorithmic output if proposed modifications cannot be agreed.

# 5 Voting Rights Act Compliance

## 5.1 The Callais Disentanglement Problem

Any analysis of algorithmic redistricting and the Voting Rights Act must begin with the core tension identified in *Louisiana v. Callais* (2025) and the line of cases addressing the interaction between VRA compliance and partisan gerrymandering.[15] The problem is epistemic: when a majority-minority district is drawn, it is often impossible to determine whether the district's shape was driven by race (potentially required by VRA Section 2) or by partisanship (potentially impermissible under the Constitution's equal protection guarantee, see *Shaw v. Reno*).[16] Because racial and partisan voting patterns are correlated in much of the country, any algorithm that draws majority-minority districts will also affect partisan compositions.

Algorithmic redistricting resolves this disentanglement problem struc-

[15] See *Louisiana v. Callais*, — U.S. — (2025); *Allen v. Milligan*, 599 U.S. 1 (2023); *Alexander v. South Carolina State Conference of the NAACP*, 144 S. Ct. 1221 (2024).

[16] *Shaw v. Reno*, 509 U.S. 630 (1993).

turally. The baseline algorithm uses no racial or partisan data; it optimizes for population equality and compactness alone. Any racial or partisan pattern in its outputs reflects the interaction between geographic voter distribution and compactness optimization, not a conscious choice to favor any racial or partisan group. When the baseline algorithm's outputs are subjected to a VRA analysis—examining whether minority populations are packed into any single district to dilute their influence, or cracked across multiple districts to prevent any majority-minority district from forming—the analysis examines what geographic compactness alone produces, providing a clean baseline against which VRA adjustment can be measured.

The VRA mode of the algorithm then operates as a documented, transparent modification: it increases the weight on edges connecting census tracts with high minority populations, making it more expensive for the partitioner to cut through minority communities. This edge-weighting approach, described formally in **?**, creates majority-minority districts by preserving the geographic clustering of minority populations—it does not "target" any specific percentage of minority voters but instead prevents the algorithm from fragmenting natural geographic concentrations. Because the modification is mathematical and documented, courts can examine precisely what the edge-weighting did and did not do. The disentanglement problem that plagued *Callais* does not arise because the record contains a baseline (no VRA mode), a modification (VRA-weighted mode), and a documented explanation of exactly what data drove each.

## 5.2 Section 2 and the *Gingles* Framework

Section 2 of the Voting Rights Act prohibits voting practices or procedures that result in the denial or abridgement of the right to vote on account of race, color, or membership in a language minority group.[17] Under *Thornburg v. Gingles* (1986), a Section 2 vote dilution claim based on single-member district lines requires proof of three preconditions:[18] (1) the minority group is "sufficiently large and geographically compact to constitute a majority" in a single-member district; (2) the minority group is "politically cohesive"; and (3) the majority votes "sufficiently as a bloc to enable it ... usually to defeat the minority's preferred candidate."

Algorithmic redistricting operates cleanly within this framework. Prong 1 (geographic compactness) is directly served by the algorithm's compactness optimization: the algorithm is less likely than a human mapmaker to create artificial geographic configurations that spread a geographically compact minority population across multiple districts. Empirical evidence supports this: the algorithm analyzed in **?** produces 69 more majority-minority districts than the enacted plans for the same states, demonstrating that compactness-optimizing redistricting serves rather than undermines Section 2's geographic-compactness requirement.

Prongs 2 and 3 (political cohesion and majority bloc voting) are not affected by the algorithm at all—they are empirical questions about voter

[17] 52 U.S.C. § 10301.

[18] *Thornburg v. Gingles*, 478 U.S. 30, 50-51 (1986).

behavior that neither the algorithm nor the redistricting process determines. If a community of color is politically cohesive and faces majority bloc voting sufficient to defeat its preferred candidates, these facts exist in the world independent of how district lines are drawn. The algorithm's contribution is to ensure that district lines do not artificially prevent the satisfaction of prong 1 through non-compact configurations that dilute geographically concentrated minority populations.

## 5.3 The VRA Safe Harbor

The model statute includes a safe harbor provision for majority-minority districts produced by the VRA mode of the algorithm. States that run the algorithm in VRA mode, document the edge-weighting parameters, and produce districts that satisfy the *Gingles* preconditions for identified minority communities shall be presumed to have satisfied Section 2, with the burden shifting to any challenger to demonstrate that an alternative configuration would produce additional Section 2 districts without violating any other statutory criterion.

This safe harbor addresses two concerns simultaneously. It protects states from Section 2 liability when the algorithm's VRA mode produces majority-minority districts consistent with geographic concentration of minority populations. And it protects those majority-minority districts from *Shaw v. Reno* racial gerrymandering challenges by establishing that the district's shape was driven by documented mathematical parameters applied to geographic clustering data, not by a conscious effort to maximize any particular racial percentage.

The *Shaw* doctrine's concern is with districts whose "bizarre" shapes can be explained only by racial motivation.[19] Districts produced by a compactness-optimizing algorithm with documented VRA-mode edge-weighting have a complete non-racial explanation for every aspect of their shape: compactness optimization explains the general configuration, and geographic minority-population clustering explains why the VRA-mode adjustment preserved particular boundaries. The racial motivation concern simply does not apply to a process that can explain every boundary decision in terms of documented, non-racial optimization parameters.

## 5.4 Post-*Allen v. Milligan* Landscape

*Allen v. Milligan* (2023) reaffirmed the Section 2 framework and rejected Alabama's argument that race-conscious redistricting is unconstitutional whenever an alternative race-neutral plan would produce fewer majority-minority districts.[20] The Court held that Section 2 sometimes requires majority-minority districts and that the state must demonstrate that its plan does not result in minority vote dilution under the *Gingles* framework.

Algorithmic redistricting is well-positioned under *Allen*. Because the algorithm's baseline mode produces no majority-minority districts through any

[19] *Shaw v. Reno*, 509 U.S. at 644.
[20] *Allen v. Milligan*, 599 U.S. 1 (2023).

intentional design, any majority-minority districts in the baseline output reflect natural geographic concentration of minority populations—they satisfy the *Gingles* compactness prong by construction. The VRA mode produces additional majority-minority districts where the baseline might have fragmented geographically concentrated populations. The net result, documented in **?**, is that the algorithm produces more majority-minority districts than enacted plans without ever targeting any specific racial percentage, precisely because it does not fragment naturally compact minority communities.

*Alexander v. South Carolina* (2024) added a new wrinkle by raising the evidentiary standard for proving racial gerrymandering in the context of districts where race and partisanship are correlated.[21] Algorithmic redistricting addresses this concern directly: the algorithm produces a documented record of exactly what data influenced each boundary decision. Where race and partisanship are correlated, the algorithm's record can demonstrate which factor—if either—drove any particular boundary. In most cases, the answer will be neither: the boundary will be driven by the compactness optimization of geographic adjacency, which is independent of both race and partisanship.

# 6 Anticipated Legal Challenges

Algorithmic redistricting will face legal challenges from political actors who benefit from the current system. This section addresses the most plausible

[21] *Alexander v. South Carolina State Conference of the NAACP*, 144 S. Ct. 1221 (2024).

challenges and explains why each fails.

## 6.1 Nondelegation

*Challenge:* Congress cannot delegate the power to draw congressional districts to a computer algorithm. The nondelegation doctrine requires Congress to provide an "intelligible principle" to guide any delegation of legislative power.[22] An algorithm is too complex and insufficiently supervised to satisfy this requirement.

*Why this fails:* The nondelegation argument mischaracterizes what algorithmic redistricting does. Congress does not delegate legislative power to the algorithm; it specifies, by statute, the criteria (population equality, compactness, contiguity, no partisan data) and directs executive agencies to implement those criteria using the certified algorithmic method. This is delegation to the executive branch—the Census Bureau—with the algorithm as the Bureau's implementation tool. Congress delegating to an agency with an intelligible principle is constitutionally uncontroversial; the agency using software to implement the delegation raises no additional constitutional question.[23]

The intelligible principle for algorithmic redistricting is, in any event, far more specific than most delegations the Court has upheld. "Produce congressional districts that equalize population within $\pm 0.5\%$, minimize boundary

[22] See *J.W. Hampton, Jr. & Co. v. United States*, 276 U.S. 394, 409 (1928).

[23] See *Whitman v. American Trucking Ass'ns*, 531 U.S. 457 (2001) (upholding delegation with intelligible principle).

perimeter relative to district area, maintain contiguity, and use no data on partisan affiliation or voting history" is more precise than "protect public health" (upholding EPA air quality standards), "prevent unfair practices" (upholding FTC authority), or "prevent excessive rates" (upholding rate regulation authority).[24]

The correct analogy is the Huntington-Hill apportionment statute. Congress mandated that the Census Bureau compute apportionment using a specific mathematical formula. No court has ever suggested this is a nondelegation problem. The formula is the congressional specification; the Bureau's computation is executive implementation. The algorithmic redistricting statute follows the same pattern.

*Nondelegation under* Gundy v. United States. Four current justices in *Gundy v. United States*, 588 U.S. 128 (2019), indicated readiness to revisit the intelligible-principle doctrine and impose a more demanding standard.[25] For DIA purposes, the intelligible principle is: "minimise edge-cut in the census-tract adjacency graph subject to population balance, contiguity, and VRA compliance." This is more specific than any contested delegation that has reached the Court—the algorithm itself *is* the regulation, leaving no discretionary gap for an agency to fill through policymaking judgment. The concern that animated the *Gundy* dissenters—that open-ended delegations

[24] See *Yakus v. United States*, 321 U.S. 414 (1944); *American Power & Light Co. v. SEC*, 329 U.S. 90 (1946).

[25] See *Gundy*, 588 U.S. at 161–62 (Gorsuch, J., dissenting, joined by Roberts, C.J., and Thomas, J.); *Paul v. United States*, 140 S. Ct. 342 (2019) (Kavanaugh, J., respecting denial of certiorari) (expressing interest in revisiting the doctrine).

transfer legislative power from Congress to unaccountable agencies—does not apply where Congress has specified the precise mathematical procedure the agency must execute. A statute that tells the Bureau "compute districts using these mathematical criteria" differs categorically from delegations like "regulate in the public interest" because the Bureau exercises no policy discretion: it runs a deterministic computation.

## 6.2 Elections Clause Limits

*Challenge:* The Elections Clause grants Congress authority to regulate the "manner" of elections, but drawing district boundaries is not the "manner" of an election—it is a predicate to elections that establishes the electoral constituencies. Therefore, Congress lacks authority to mandate how states draw districts.

*Why this fails:* This argument was resolved in *Smiley v. Holm* (1932). The Court held that the Elections Clause's reference to "manner" of elections encompasses the drawing of congressional district boundaries: "The subject of the time, place and manner of elections ... necessarily includes the authority to require that congressional elections shall be held in particular districts."[26] The Court sustained a federal statutory requirement that congressional elections be held in single-member districts as a regulation of the "manner" of elections within the scope of Congress's Clause authority.

*Cook v. Gralike* (2001), while overturning Missouri's congressional ballot

[26] *Smiley v. Holm*, 285 U.S. at 366.

rotation statute as inconsistent with the Elections Clause, confirmed that the Clause grants Congress broad authority to regulate "how" elections are held.[27] A mandate specifying the method by which congressional districts are produced is squarely within this "how" authority.

## 6.3 VRA Conflict: Algorithm Produces Too Few Majority-Minority Districts

*Challenge:* The algorithm's baseline mode does not intentionally create majority-minority districts and may produce fewer such districts than enacted plans that have been designed to satisfy VRA requirements. Adopting algorithmic redistricting would violate Section 2.

*Why this fails:* Empirical evidence refutes the factual premise. The algorithm produces more majority-minority districts than enacted plans on average—69 more across the 50-state analysis in **?**—precisely because it does not artificially fragment geographically concentrated minority populations for partisan purposes. Enacted plans in states with Republican-controlled legislatures have frequently drawn minority voters into fewer majority-minority districts than compactness optimization would produce. This is the violation that *Allen v. Milligan* addressed; algorithmic redistricting runs in the opposite direction.

Moreover, the VRA does not require intentional creation of majority-minority districts—it prohibits dilution. An algorithm that produces majority-

[27] *Cook v. Gralike*, 531 U.S. 510 (2001).

minority districts as a natural consequence of compactness optimization satisfies Section 2 even if it does not target any specific racial percentage. The VRA mode further ensures that wherever a geographically compact minority community exists and satisfies the *Gingles* preconditions, the algorithm will not fragment it.

### 6.4 VRA Conflict: Algorithm Produces Racially Gerrymandered Districts

*Challenge:* The algorithm's VRA mode, which increases edge weights on census tracts with high minority populations, uses race as a factor in drawing districts. Under *Shaw v. Reno* and its progeny, this is presumptively unconstitutional racial gerrymandering.

*Why this fails:* *Shaw* targets districts whose shapes are so "bizarre" that they are "unexplainable on grounds other than race."[28] The VRA-mode algorithm has a complete non-racial explanation for every boundary it draws: it is minimizing the edge cut of a graph in which certain edges have been assigned higher weights based on the demographic concentration of the census tracts they connect. The result of this optimization is that geographically compact minority communities are preserved—the same outcome that would result from a human mapmaker who chose not to fragment natural communities. The "bizarreness" concern does not apply to mathematically compact districts that can be explained by geography.

[28] *Shaw v. Reno*, 509 U.S. at 644.

Furthermore, using race as a factor to satisfy VRA Section 2 is constitutionally permissible provided it is "narrowly tailored" to achieving compliance and does not go beyond what compliance requires.[29] The VRA mode applies edge-weighting to geographic adjacency—it does not target any population percentage threshold but instead prevents fragmentation of geographic minority concentrations. This is a narrow use of demographic information: it preserves what geography created rather than engineering what race requires.

*Race-predominance under* Miller v. Johnson. *Shaw* doctrine has evolved through *Miller v. Johnson*, 515 U.S. 900 (1995), and *Bethune-Hill v. Virginia State Board of Elections*, 580 U.S. 178 (2017), to focus on whether racial considerations *predominated* over traditional redistricting principles, not merely whether district shapes are bizarre. Under the predominant-factor test, a court would ask whether race overrode the algorithm's compactness objective. Under the DIA, no racial data enters the partitioning algorithm. Any VRA adjustment is separately documented under § 2(e) and is analytically separable from the base partition: the base partition is produced by the certified algorithm using no demographic data, and the VRA adjustment modifies only specific districts for which a written Gingles justification has been published. Because race modifies the compactness optimization at the margin rather than replacing it, race does not predominate over traditional redistricting criteria—compactness remains the governing principle, with VRA compliance as a documented statutory constraint, in the same way that pop-

[29] See *Bush v. Vera*, 517 U.S. 952 (1996).

ulation equality is a constraint rather than the "predominant factor" in every redistricting plan.

## 6.5 Communities of Interest

*Challenge:* Many state constitutions require redistricting bodies to consider communities of interest, including economic, social, and political communities that may not align with geographic compactness. An algorithm that optimizes solely for compactness cannot satisfy this requirement.

*Why this fails:* This challenge conflates what the algorithm produces by default with what the full statutory scheme permits. The model statute provides an adjustment process in which communities of interest may be identified by local advocates, municipal governments, and regional planning bodies. The algorithm's baseline is the starting point, not the final answer. Proposed adjustments that preserve a documented community of interest—a Native American tribal land, a port city's waterfront community, a historically contiguous immigrant neighborhood—are permissible under the statute provided they do not reduce compactness below the statutory threshold and maintain population balance.

More fundamentally, compact districts tend to preserve communities of interest better than non-compact districts. Research by **?** shows that 80% of algorithmic district boundaries are geographically stable across three census cycles—indicating that the districts reflect underlying community geography rather than arbitrary line-drawing. Districts that track the underlying geography of a state tend to align with communities that have organized around that geography.

## 7 Model Statute

The following model statute is drafted in U.S. Code style for purposes of federal enactment. State legislatures may adapt this text by substituting the relevant state agency for the Census Bureau, the relevant state redistricting body for the redistricting commission, and adding any state-specific adjustments required by the state constitution's substantive redistricting criteria.

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**THE DISTRICTING INTEGRITY ACT**

*An Act to establish a uniform algorithmic method for congressional redistricting and to vest implementation authority in the Bureau of the Census. Be it enacted by the Senate and House of Representatives of the United States of America in Congress assembled,*

**SECTION 1. FINDINGS AND PURPOSE.**

**(a) Findings.** Congress finds that—

(1) Congressional redistricting by partisan state legislatures has produced district maps designed to entrench political parties and insulate incumbents from electoral accountability, undermining the constitutional

guarantee that the House of Representatives shall be chosen "by the People."

(2) The Supreme Court held in *Rucho v. Common Cause*, 588 U.S. 684 (2019), that federal courts lack judicially manageable standards for adjudicating partisan gerrymandering claims, and identified federal legislation as an appropriate remedy.

(3) Congress has authority under Article I, Section 4 of the Constitution to regulate the manner of congressional elections, including the method by which congressional districts are drawn, as confirmed in *Smiley v. Holm*, 285 U.S. 355 (1932).

(4) Mathematical algorithms that optimize for population equality and geographic compactness using only data from the decennial census are structurally incapable of incorporating partisan data and therefore structurally immune to partisan manipulation.

(5) The method of equal proportions (Huntington-Hill method), codified at section 2a of this title, demonstrates that Congress may successfully mandate mathematical methods for politically sensitive calculations, producing legitimate and stable outcomes that have operated without controversy since 1941.

(6) Algorithmic redistricting satisfies the requirements of section 2 of the Voting Rights Act of 1965 (52 U.S.C. § 10301) and produces on average

more majority-minority districts than legislatively enacted plans.

(7) Public confidence in the fairness of congressional redistricting is a legitimate governmental interest that algorithmic redistricting advances by providing transparent, reproducible, and auditable district maps.

**(b) Purpose.** The purpose of this Act is to establish an objective, transparent, and manipulation-resistant method for congressional redistricting that satisfies constitutional population-equality requirements, produces geographically compact and contiguous districts, and eliminates structural opportunities for partisan manipulation of district boundaries.

## SECTION 2. ALGORITHMIC REDISTRICTING METHOD.

**(a) Definitions.** For purposes of this section—

(1) The term "certified redistricting algorithm" means a computer program that—

- (A) uses only data from the most recent decennial census conducted pursuant to section 141 of title 13 and geographic boundary data maintained by the Bureau under the Topologically Integrated Geographic Encoding and Referencing (TIGER) system;
- (B) does not use, directly or indirectly, any data on the partisan affiliation, voting history, or electoral preferences of any person or geographic unit;
- (C) produces congressional district maps that—

(i) achieve equal population among districts, with maximum deviation not to exceed one-half of one percent of the average district population;

(ii) ensure that all districts are geographically contiguous;

(iii) minimize the ratio of district boundary perimeter to district area (maximize compactness) subject to the population-equality constraint; and

(iv) do not divide census tracts except where necessary to satisfy population-equality requirements;

(D) produces identical output for identical input data; and

(E) has source code that is published and publicly accessible.

(2) The term "preliminary district map" means the congressional district map produced by the certified redistricting algorithm for a State without any human modification.

(3) The term "final district map" means the congressional district map adopted for a State under subsection (d).

**(b) Certification of Algorithm.** Not later than 180 days after the date of enactment of this Act, the Bureau of the Census shall—

(1) certify a redistricting algorithm that satisfies the requirements of subsection (a)(1);

(2) publish the certified algorithm's source code on a publicly accessible Federal website; and

(3) promulgate regulations establishing procedures for running the certified algorithm, reviewing its outputs, and requesting adjustments.

**(c) Production of Preliminary District Maps.** Not later than 30 days after the Bureau transmits the apportionment statement required by section 2a(a) of this title, the Bureau shall—

(1) run the certified redistricting algorithm for each State;

(2) produce a preliminary district map for each State;

(3) publish each preliminary district map and the underlying data and parameters on a publicly accessible Federal website; and

(4) transmit each preliminary district map to the Governor and presiding officer of each chamber of the legislature of the State for which the map was produced.

**(d) Adjustment Period and Final Map Adoption.**

(1) *Comment period.* For a period of 60 days following the publication of the preliminary district map for a State, any person may submit to the Bureau proposed adjustments to the preliminary district map. Proposed adjustments must—

(A) be documented by reference to objective geographic, municipal, or community-of-interest factors;

(B) not introduce any data on the partisan affiliation, voting history, or electoral preferences of any person or geographic unit;

(C) maintain population equality within the limits specified in subsection (a)(1)(C)(i); and

(D) not reduce the compactness of any district more than 5 percent below the compactness of the corresponding district in the preliminary district map.

(2) *Bureau review.* Not later than 30 days after the close of the comment period, the Bureau shall—

(A) review all proposed adjustments for compliance with paragraph (1);

(B) accept, modify, or reject each proposed adjustment, with written explanation; and

(C) publish the final district map for each State.

(3) *Default rule.* If no timely adjustments are received, or if all adjustments are rejected, the preliminary district map shall serve as the final district map.

**(e) Voting Rights Act Adjustment.** Notwithstanding subsection (b), a State may deviate from the certified redistricting algorithm's output to the extent necessary to comply with Section 2 of the Voting Rights Act of 1965 (52 U.S.C. § 10301), provided that—

(1) the deviation is limited to the minimum necessary to achieve compliance with Section 2;

(2) the State publishes a written justification documenting satisfaction of the preconditions established in *Thornburg v. Gingles*, 478 U.S. 30 (1986), for each minority community affected by the deviation;

(3) partisan affiliation data is not used in determining the adjustment, consistent with *Louisiana v. Callais*, 608 U.S. __ (2026); and

(4) the adjusted district map and the supporting Gingles justification are published on the same publicly accessible Federal website as the preliminary district map under subsection (c)(3) not later than 15 days before the final map adoption deadline under subsection (d)(2).

## SECTION 3. JUDICIAL REVIEW AND ENFORCEMENT.

**(a) Jurisdiction.** A three-judge district court convened pursuant to section 2284 of title 28 shall have original jurisdiction over any action challenging—

(1) the certification of the redistricting algorithm under section 2(b);

(2) the preliminary or final district map produced for any State; or

(3) any adjustment accepted or rejected by the Bureau under section 2(d).

**(b) Grounds for Challenge.** A challenge to a final district map shall be limited to claims that—

(1) the certified algorithm failed to satisfy the requirements of section 2(a)(1);

(2) the Bureau failed to run the certified algorithm in accordance with the regulations promulgated under section 2(b)(3);

(3) the final district map fails to satisfy the population-equality, contiguity, or compactness requirements of section 2(a)(1)(C); or

(4) an accepted adjustment introduced data on partisan affiliation, voting history, or electoral preferences in violation of section 2(a)(1)(B).

**(c) Time Limits.** Any action under this section shall be filed not later than 30 days after the publication of the final district map. The three-judge district court shall issue a final decision not later than 60 days after the filing of the complaint.

**(d) Appeals.** Appeal from a decision of the three-judge district court under this section shall lie directly to the Supreme Court pursuant to section 1253 of title 28.

**(e) Effective Date.** This Act shall take effect with respect to redistricting conducted following the first decennial census conducted after the date of enactment.

## 7.1 Drafting Notes

Several features of this model statute deserve explanation. First, the operative mandate is stated minimally: the statute specifies required properties of the algorithm (what it must use, what it must not use, what it must produce) without prescribing any particular algorithmic implementation.

*Note on compactness measurement.* Section 2(a)(1)(C)(iii) defines compactness as minimizing "the ratio of district boundary perimeter to district area"—the Polsby-Popper formulation. The reference implementation described in **?** instead minimizes edge cut in the census-tract adjacency graph, which correlates with but is not identical to Polsby-Popper score. These two compactness criteria are related: districts with short borders in geographic space tend to cross fewer census-tract boundaries. For statutory purposes, we define "maximum compactness" as the minimum-edge-cut partition of the TIGER adjacency graph—an objective criterion independent of shapefile-specific perimeter measurement. Courts challenging a map on compactness grounds would evaluate whether the certified algorithm was run correctly on the published adjacency graph, not whether the Polsby-Popper score exceeds a specific threshold. State legislatures that prefer a Polsby-Popper standard may substitute "minimize the ratio of district boundary perimeter to district area" for "minimize graph edge-cut" in subsection (a)(1)(C)(iii), with the tradeoff that Polsby-Popper scores are sensitive to coastline and river-boundary complexity in ways that graph edge-cut is not.

The overall property-specification approach is intentional—it allows the Bureau to certify the best available algorithm at the time of implementation and to recertify improved algorithms as computer science advances, without requiring statutory amendment. The analogy is the Bureau's authority to conduct the census using updated statistical methodology; the statute specifies the census's purpose and output requirements, not its technical implementation.

Second, the adjustment provision in Section 2(d) is deliberately narrow. It permits adjustments based on documented geographic and community factors but explicitly prohibits the introduction of partisan data at the adjustment stage. This prevents the adjustment process from becoming a backdoor for the partisan manipulation the statute is designed to eliminate. The 5% compactness floor ensures that adjustments cannot substantially degrade the algorithm's outputs.

Third, the judicial review provision in Section 3 is administratively manageable—a deliberate contrast to the partisan gerrymandering standard that *Rucho* found unmanageable. Courts review whether the algorithm was certified correctly, whether it was run correctly, and whether the outputs meet the statutory technical specifications. These are questions courts can answer without engaging in the normative balancing that defeated judicial oversight of partisan gerrymandering.

# 8 Conclusion

The legal case for algorithmic redistricting is strong on every dimension examined in this Article. The constitutional authority exists—the Elections Clause, as construed in *Smiley v. Holm*, permits Congress to mandate the method of congressional redistricting, and the Huntington-Hill statute provides a direct structural precedent for mathematical mandates in the apportionment-redistricting domain. The state adoption pathways are clear—42 states can adopt algorithmic redistricting by ordinary statute or commission resolution without constitutional amendment, and the remaining eight face modest amendment requirements. The VRA compliance case is strong—the algorithm produces more majority-minority districts than enacted plans, and its VRA mode creates majority-minority districts through documented compactness optimization rather than racial targeting, satisfying the *Gingles* framework while avoiding the *Shaw* racial gerrymandering concern. And the anticipated legal challenges do not hold up: the nondelegation claim mischaracterizes delegation to an executive agency implementing a statutory mandate as delegation to the algorithm itself; the Elections Clause limits claim was resolved in 1932; and the VRA conflict claim gets the empirical facts backwards.

The model statute in Section **??** provides ready-to-use text for federal or state enactment. Its three-section structure—definitions and certification, map production and adjustment, judicial review—can be adapted to any state's existing redistricting statutory scheme with minimal modification. State legislatures working from this template should be attentive to two adaptation requirements: substituting the appropriate state agency for the Bureau of the Census, and ensuring that the adjustment criteria are consistent with any constitutional redistricting standards their state constitution specifies.

The post-*Rucho* moment is, paradoxically, an opportunity. The federal courthouse is closed to partisan gerrymandering claims. State courts are active but inconsistent. Congress has an explicit invitation from the Chief Justice to act. The tool is ready, the legal authority is clear, and the public supports reform.[30] The question is whether the political will exists to use them.

The experience of the Huntington-Hill apportionment method should be instructive. Before 1941, apportionment was among the most contentious recurring political disputes in American governance. After 1941, the mathematical formula rendered the question technical, predictable, and uncontroversial. Eighty years of operational success is strong evidence that mathematical mandates can resolve politically charged democratic governance questions—not by choosing winners and losers, but by ensuring that the process of choosing cannot itself be weaponized. Congressional redistricting is overdue for the same resolution.

[30]See **?** (finding 71% of survey respondents support their state using computer algorithms to draw congressional districts).

*The author thanks the algorithmic redistricting research team for technical collaboration and data access. All errors are the author's own.*

# References

Giovanni Della-Libera. From apportionment to boundary design: Extending huntington-hill to congressional redistricting. *Working paper*, 2026a.

Giovanni Della-Libera. Temporal stability of algorithmic congressional districts across census cycles. *Working paper*, 2026b.

Giovanni Della-Libera. Public perceptions of algorithmic vs. human-drawn congressional districts: A survey experiment. *Working paper*, 2026c.

Giovanni Della-Libera. Algorithmic redistricting and voting rights act compliance: Compactness optimization produces more majority-minority districts. *Working paper*, 2026d.